

Anastasia Galperina

Computational Biologist | Bioinformatician | Translational Research

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Data scientist with experience in developing data pipelines, multi-omic integration, and clinical data analysis for biomarker discovery

TECHNICAL SKILLS

Bioinformatics & Computational Biology :

- NGS Analysis: RNA-Seq, WGS metagenomics, genome assembly (de novo & reference-based), WGCNA
- Multi-omics Integration: Network analysis, factor analysis, proteomics, data integration
- Clinical Data: Big data management, statistical analysis, proteomic biomarker identification

Programming & Tools:

- Languages: R, Python, SQL
 - Environments: Linux/Unix, HPC systems, Git, Bioconductor, BioConda, RShiny, Spotfire
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PROFESSIONAL EXPERIENCE

GSK (Cambridge, MA)

Digitization Business Analyst | Sept 2024 – Current

- Developed automated data visualization and analysis pipeline for lab equipment, ensuring ALCOA+ compliance and data integrity.
- Proposed, planned, executed, and assisted in managing projects, such as software implementation, and data collection and analysis projects.
- Led cross-functional collaboration across three countries, supporting 100+ scientists with data digitization needs.
- Designed and delivered training programs for novel data-capturing software for over 100 scientists.
- Optimized and managed SQL databases, ensuring efficient data storage, retrieval, and integration of scientific applications.

University of Helsinki (Helsinki, FI)

Research Assistant, Human Microbiome Research Program | Jan 2023 – March 2024

- Engineered comprehensive viral database (145,000+ entries) via systematically surveying 9 previously published human gut viral sources and assessing their overlap and quality.
- Data analysis of 1900 metagenomic samples from a longitudinal clinical study of infants and investigated the prevalence and dynamics of viral communities during development.
- Assembled, measured quality and completeness, and predicted gene annotations of bacterial genomes.
- Produced reproducible R Markdown reports identifying technical biases from different sequencing approaches to assist with methodology decision making.
- Contributed to scientific conference presentations and translational publications.

AbbVie (Worcester, MA)

Omics Bioinformatician Intern | June 2023 – Aug 2023

- Determined an optimized analytical machine learning strategy for multi-omic analysis and convergence with the overarching goal of identifying biomarker targets for the treatment of IBD.
- Developed scalable RNA-Seq data processing workflow specialized for machine learning analysis and multi-omic integration.
- Presented findings at departmental meetings to assist with planning of new experiments with internal and external partners.

Biologist I | Sept 2021 – Jan 2022 and May 2022 – Aug 2022

- Conducted differential expression analysis on patient data from over 50 drug development clinical trial studies.
 - Maintained and automated bioinformatic pipelines for mass spectrometry proteomic data.
 - Implemented standardized protocols for protein identification in complex biological samples, improving reproducibility across multiple research projects.
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Education

2024	Translational Medicine MSc	University of Helsinki, Finland
2022	Biology B.S.; Computer Science B.A.	University of Massachusetts, Amherst

Publications

Galperina A.; Lugli G.; Milani C.; De Vos W.; Ventura M.; Salonen A.; Hurwitz B.; Ponsero A.J. The Aggregated Gut Viral Catalogue (AVrC): A Unified Resource for Exploring the Viral Diversity of the Human Gut. *In revision* PLOS computational biology.